Rishabh Agarwal

Senior Frontend Engineer · React · TypeScript · Frontend System Design

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SUMMARY

Senior Frontend Engineer with 7+ years building scalable, high-performance and accessible web applications for healthcare, banking and SaaS products with React, Angular and TypeScript. Owns work end to end — architecture, performance, security and delivery — and recently designed, built and launched **NamasteScale**, a full-stack SaaS (React 19, NestJS, PostgreSQL, AWS), solo from first commit to production.

EXPERIENCE

Senior Associate II — Alcon Laboratories — Oct 2024 – Present · Bengaluru

- Build the **Marlo** healthcare platform (B2B2C and B2C), delivering Patient and Eye Care Professional (ECP) portals for end-to-end care workflows.
- Built the ECP portal with React, Redux Toolkit and Material UI, **reducing state-management complexity by 40%**, improving UI consistency and integrating OAuth and Sequra services.
- Optimized the Angular/TypeScript Patient portal with a modular, responsive architecture, **improving page-load performance by 30%** and speeding up feature delivery.

Senior Frontend Developer — Vorro India — Feb 2023 – Oct 2024 · Remote

- Developed a React data-transformation platform, **improving processing performance by 40%** and **data accuracy by 25%** through optimized workflows and validation logic.
- Designed a reusable component architecture for complex, dynamic states, **increasing stability by 30%** and **cutting development effort by 20%**.

Consultant — Deloitte — Jan 2022 – Jan 2023 · Gurugram

- Led frontend development of a waste-management tracking application (Angular, React, TypeScript), **reducing scheduling errors by 20%** through accurate driver-hour tracking and **raising user satisfaction by 25%**.
- Streamlined time-tracking and reporting workflows, **improving operational efficiency by 30%**.

Senior Software Engineer — Capgemini — Mar 2019 – Jan 2022 · Pune

- Designed and delivered **Genie**, a secure enterprise search and document-management platform for banking users, unifying database and file access with **role-based access control** for compliant sharing.
- Delivered AIQUIC and FNOL integrations, **reducing analysis time by 30%** and **settlement time by 40%** with **95% processing accuracy**.

PROJECTS

NamasteScale — Founder & sole engineer · namastescale.com — Launched Nov 2026

- Built and launched a system-design learning SaaS: React 19 app (TanStack Query, Zustand, React Flow), NestJS 11/Fastify API with a BullMQ worker, PostgreSQL (Drizzle, 30+ tables) and Redis — **120+ versioned REST endpoints**.
- Shipped AI reviewer, mentor, mock interviewer and AI-assisted debugging round on Claude with rate limits, plan-based model tiers and cost metering; **rule-based fallbacks** keep features working when the model is down.
- First-party auth and billing: hashed server sessions, CSRF, TOTP 2FA, Google/GitHub OIDC and **signature-verified, idempotent Razorpay webhooks**; Zod schemas shared by API and UI.
- Runs on AWS (EC2, RDS, S3, SES) with Docker and Caddy behind Cloudflare at **~80% lower cost** than the autoscaled design, with an AWS CDK path to ECS Fargate and Aurora.

react-virtual-renderer — open-source React virtualization library (npm) — react-virtual-renderer.vercel.app

- Framework-agnostic list/grid/masonry engine on a **Fenwick tree**: dynamic-height re-measurement O(n) → O(log n), **~2000× faster at 50,000 items**; masonry appends **~667× faster**.

SKILLS

Languages	JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Frontend	React, Next.js, Angular, Redux Toolkit, RTK Query, TanStack Query, Zustand, RxJS, Tailwind CSS, Material UI
Backend & Data	Node.js, Express, NestJS, REST API design, PostgreSQL, MySQL, MongoDB, Redis, BullMQ, OAuth/OIDC, JWT
Cloud & Tools	AWS (EC2, S3, CloudFront, Lambda, API Gateway, RDS, CDK), Docker, Cloudflare, GitHub Actions, Vite, Webpack, Jest, Vitest, RTL
Practices	Frontend system design, web performance, accessibility, design systems, Agile/Scrum

EDUCATION

B.Tech, Computer Science — BBD University, Lucknow · GPA 3.7/4.0, First Class with Distinction — 2018